front matter

foreword

Born as an alternative to EJBs in the early 2000s, the Spring framework quickly overtook its opponent with the simplicity of its programming model, the variety of its features, and its third-party library integrations. The Spring ecosystem grew over the years into the broadest and most mature development framework available in any programming language. Its main competitor quit the race when Oracle stopped the evolution of Java EE 8, and the community took over its maintenance via Jakarta EE.

According to recent surveys (**http://mng.bz/l9VB** and **http://mng.bz/B1Ar**), Spring is the framework underlying more than half of the Java applications. This fact builds up an enormous codebase that makes it critical for any Java developer to learn Spring, as it's inevitable you'll encounter this technology in your career. I've been developing applications with Spring for 15 years, and today the teams that I train in hundreds of companies are almost all using Spring.

The reality is that despite being so popular, it's pretty hard to find quality introductory material. The reference documentation is thousands of pages long, describing all the subtleties and details that could be helpful in very specific scenarios, so it's not an option for a newcomer. While online videos and tutorials typically fail to engage the student, very few books capture the essence of Spring framework, often spending long pages debating topics that prove to be irrelevant to the problems faced in modern application development. With this book, however, it's very hard to find anything to remove; all the concepts covered are recurring topics in the development of any Spring application.

The reader is gently brought to a level sufficient to become rapidly productive in a project based on the Spring framework. My own experience training thousands of employees showed me that the vast majority of developers working with Spring today don't see the ideas as clearly as this book paints them. Furthermore, developers are unaware of the many pitfalls about which this book warns its readers. In my opinion, this book is a must-read for any developer starting on their first Spring project.

The attention with which Laurențiu anticipates the typical questions occurring in the reader's mind proves his extensive experience teaching Spring in class. This teaching fluency allows the author to adopt a personal, warm tone that makes this book an effortless and pleasant read. The book has a clear, sharp structure, and I really loved how complex topics were progressively revealed and explained and reiterated in subsequent chapters.

This book shines in that the reader is also introduced to fundamental concerns regarding a legacy project using the Spring framework. In an ecosystem dominated by Spring Boot, I find it very useful to sneak a peek under the hood. On the other end, the book also gently introduces the reader to last-generation technologies, like Feign clients and even reactive programming.

I wish you a pleasant reading, and never hesitate to get your hands dirty with some code whenever you think things get complicated.

—Victor Rentea

Java champion, trainer, and consultant

preface

Sharing knowledge and creating learning material is a hobby for me. In addition to being a software developer, I'm also a teacher. As a Java trainer since 2009, I've taught Java to thousands of developers with various levels of experience, from university students to experienced developers in large corporations. In the past few years, I've come to consider Spring a must-learn for beginners. Apps today are no longer implemented with vanilla languages—almost everything relies on frameworks. Since Spring is the most popular application framework in the Java world today, Spring is something a developer needs to learn in their first software development steps.

In teaching Spring to beginners, I've realized that it is still treated as something you learn only when you already have some experience coding. When I started writing *Spring Start Here*, there were already plenty of tutorials, books, and articles on the topic, but my students continued to tell me they found those materials hard to understand. I realized the problem was not that the existing learning material wasn't excellent, but that there was no dedicated study guide for an absolute beginner, so I decided to write a book that doesn't consider Spring something you learn after you have some experience, but instead something you can learn with minimal foundational knowledge.

Technology changes quickly. But it's not only the technology changing. We also need to consider how we can improve the way we teach these technologies. Some years ago, one would start learning the language fundamentals and get employed as a developer without even knowing what a framework is. But today, these things are different. Learning all the details of a language up-front is no longer the way to quickly develop the skills you need to work in a software development team. Now, I recommend developers start with the fundamentals and, once they feel comfortable with the basics, start learning an application framework. Spring is, in my opinion, the best application framework to start learning. Understanding the Spring basics also opens doors to learning other technologies and changes the old, linear learning approach into something that looks more like a tree—and each branch of the tree is a new framework you learn in parallel with others.

I designed *Spring Start Here* to be the book you want to start learning the Spring framework with. This book leads you step-by-step, providing you with all the essential theoretical knowledge, accompanied by examples that practically apply the discussed topics. I hope this book will bring significant value to you, the reader, and help you quickly boost your Spring knowledge and open doors for further learning.

acknowledgments

This book wouldn't be possible without the large number of smart, professional, and friendly people who helped me throughout its development process.

First, a big thank you to my wife, Daniela, who was always there for me, and whose valuable opinions, continuous support, and encouragement were a huge help to me.

I'd also like to express my gratitude and send special thanks to all the colleagues and friends who helped me from the very first table of contents and proposal with their valuable advice.

A big thank you goes to the entire Manning team for their huge help in making this book a reality. I especially want to recognize Marina Michaels, Al Scherer, and Jean-François Morin for always being incredibly supportive and professional. Your advice has brought great value to this book.

I'd like to thank my friend Ioana Göz for the drawings she created for the book. She turned my thoughts into the cartoons in the book.

I also want to express my appreciation to all the reviewers who provided such useful feedback at every step. To Alain Lompo, Aleksandr Karpenko, Andrea Carlo Granata, Andrea Paciolla, Andres Damian Sacco, Andrew Oswald, Bobby Lin, Bonnie Malec,

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Finally, a special thank you to my friends, Maria Chițu, Andreea Tudose, Florin Ciuculescu, and Daniela Ileana for advising me along the way.

about this book

Since you've opened this book, I assume you're a software developer in the Java ecosystem who found out it's useful to learn Spring. This book teaches you the Spring foundations, assuming you know nothing in the first place about frameworks and, of course, about Spring.

You'll start with what a framework is and then gradually learn the basics of Spring with applied examples. You will not only learn to use the framework's components and capabilities, but you'll also learn the essentials of what happens behind the scenes in these capabilities. Knowing how the framework operates when you use a particular component helps you design better apps, and solve problems faster.

When you finish this book, you'll have learned the following skills, which are highly relevant in implementing apps:

- Configuring and using the Spring context and dependency injection with Spring
- Designing and using aspects
- Implementing web apps
- Implementing data exchange between apps
- Persisting data
- Testing implementations

You'll find this book is valuable for the following:

- Working on an app using Spring for your job
- Succeeding in a technical interview for a Java developer role
- Obtaining your Spring certification

Even if this book's first purpose is not to prepare you for a Spring certification, I consider it a must-read before digging into details a certification exam typically requires.

Who should read this book

This book is for developers who understand basic object-oriented programming and Java concepts and want to learn Spring or refresh their Spring fundamentals knowledge. You do not need to have previous experience with any framework, but you need to understand Java because this is the language we use throughout the book's examples.

Spring is one of the most encountered technologies in Java apps and will most likely be used even more in the future. For a Java developer, this makes Spring a must-know today. Learning what I teach you in this book will help you upskill, provide you with the Spring foundation knowledge and skills you need to successfully pass a Java interview, and work on an app using Spring technologies. The book also opens doors to further study Spring details that are more complex.

How this book is organized: A roadmap

This book is divided into two parts that cover 15 chapters. We'll start our discussion (in the first part of the book) with straightforward examples to show you how to make Spring aware of your application. We'll then build examples that enable you to understand the core of any real-world Spring app. Once we finish with Spring Core basics, we'll discuss Spring Data and Spring Boot basics.

From chapter 2 to the end of this book, you'll find that theoretical aspects are accompanied by projects in which we apply the notions we discuss. I explain the code in these examples snippet by snippet. My recommendation is you build these examples with me while reading. Then, you can compare your result with my solution.

As presented in the following figure, I designed the book's chapters to be read in the given order. In chapters 2 through 5, where we discuss the Spring context, you might find the examples predominantly theoretical. For anyone with little or no experience with Spring, it's essential to start this way. Don't worry! I present the foundations in the easiest possible way, and then our examples and discussions gradually become more sophisticated to reflect real-world, production-ready code.

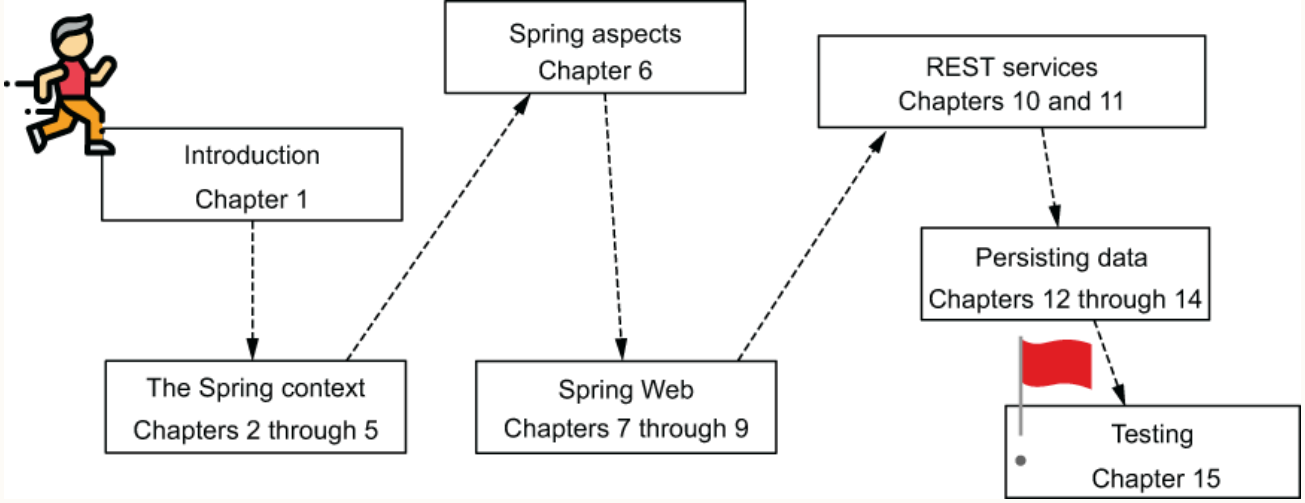


Figure 1 If you start with no (or very little) knowledge about Spring, the best way to read the book is to start with the first chapter and read everything in order.

If you already understand the Spring context and Spring AOP well, you can skip part 1 and go directly to part 2, "Implementation" (chapters 7-15), as presented in the next figure.

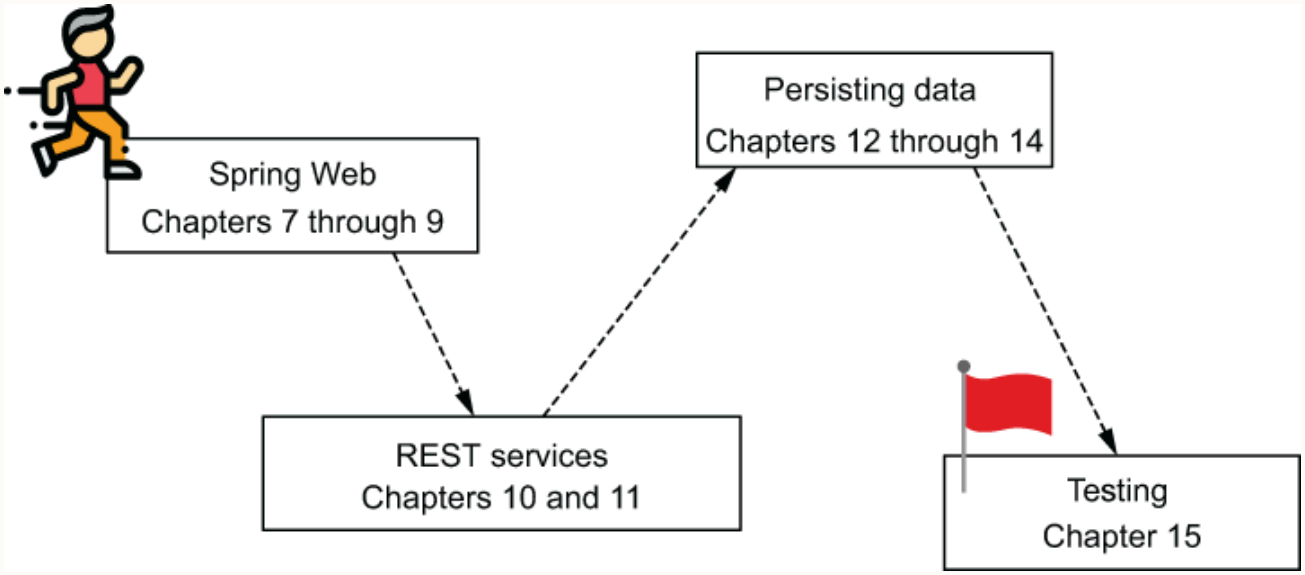


Figure 2 If you already understand the Spring framework's foundation and know how to use the Spring context and design aspects, you can start with part 2, where we use Spring capabilities to implement apps mirroring scenarios you face in real-world systems.

Once you finish reading this book, you'll have learned plenty of skills to develop apps like a professional. You'll learn to connect to databases using the most encountered techniques today, and you'll learn how to make apps communicate with each other. We'll end the book with teaching a critical topic: testing. I'll season the text here and there by adding stories of my experience and notes with valuable pieces of advice.

Remember that Spring is a vast universe, and one book won't teach you everything about it. With this book, you get started with the framework and learn the fundamental skills

of using Spring's valuable components. Throughout the book, I refer, where appropriate, to other resources and books that detail the topics we discuss. I strongly recommend you read those additional resources and books to broaden your perspective on the discussed topics.

bout the code

The book provides about 70 projects, which we'll work on in chapters 2 through 14. When working on a specific example, I mention the project's name, which implements that example. My recommendation is to try to write your example from scratch and then use the provided project only to compare your solution with mine. This approach will help you better understand the concepts you're learning.

Each of the projects is built with Maven, making it easy to import into any IDE. I used IntelliJ IDEA to write the projects, but you can choose to run them in Eclipse, Netbeans, or any other tool of your choice. Appendix F gives you an overview of the recommended tools.

This book contains many examples of source code, both in numbered listings and in line with normal text. In both cases, source code is formatted in a fixed-width font like this to separate it from ordinary text. Sometimes code is also **in bold** to highlight code that has changed from previous steps in the chapter, such as when a new feature adds to an existing line of code. In many cases, the original source code has been reformatted; we've added line breaks and reworked indentation to accommodate the available page space in the book. In rare cases, even this was not enough, and listings include line-continuation markers (↪). Additionally, comments in the source code have often been removed from the listings when the code is described in the text. Code annotations accompany many of the listings, highlighting important concepts.

veBook discussion forum

Purchase of *Spring Start Here* includes free access to a private web forum run by Manning Publications where you can make comments about the book, ask technical questions, and receive help from the author and from other users. To access the forum, go to **https://livebook.manning.com/#!/book/spring-start-here/discussion**. You can also learn more about Manning's forums and the rules of conduct at **https://livebook.manning.com/#!/discussion**.

Manning's commitment to our readers is to provide a venue where a meaningful dialogue between individual readers and between readers and the author can take place. It is not a commitment to any specific amount of participation on the part of the author, whose

contribution to the forum remains voluntary (and unpaid). We suggest you try asking the author some challenging questions lest his interest stray! The forum and the archives of previous discussions will be accessible from the publisher's website as long as the book is in print.

about the author



Laurențiu Spilcă is a dedicated development lead and trainer at Endava, where he leads the development of projects in the financial market with users in Europe, the US, and Asia. He has over 10 years of experience. Laurențiu believes it's important to not only deliver high-quality software, but also share knowledge and help others to upskill. These beliefs have driven him to design and teach courses related to Java technologies and deliver presentations and workshops. His Twitter handle is @laurspilca.

about the cover illustration

The figure on the cover of *Spring Start Here* is captioned "Femme d'ajaccio isle de Corse," or a woman from Ajaccio on the island of Corsica. The illustration is taken from a collection of dress costumes from various countries by Jacques Grasset de Saint-Sauveur (1757-1810), titled *Costumes de Différents Pays*, published in France in 1797. Each illustration is finely drawn and colored by hand. The rich variety of Grasset de Saint-Sauveur's collection reminds us vividly of how culturally apart the world's towns and regions were just 200 years ago. Isolated from each other, people spoke different dialects and languages. In the streets or in the countryside, it was easy to identify where they lived and what their trade or station in life was just by their dress.

The way we dress has changed since then and the diversity by region, so rich at the time, has faded away. It is now hard to tell apart the inhabitants of different continents, let alone different towns, regions, or countries. Perhaps we have traded cultural diversity for a more varied personal life—certainly for a more varied and fast-paced technological life.

At a time when it is hard to tell one computer book from another, Manning celebrates the inventiveness and initiative of the computer business with book covers based on the rich diversity of regional life of two centuries ago, brought back to life by Grasset de Saint-Sauveur's pictures.

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